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For whom do we densify? Explaining income variation across densification projects in the region of Utrecht, the Netherlands

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Abstract

While governments worldwide rely upon compact city policies to reduce land consumption from urban growth, recent studies have addressed the potential trade-off between densification and housing affordability. Concerns have been voiced that densification leads to a one-sided housing supply, structurally excluding low-income households. However, few studies address household income variation across densification projects, leaving us with a limited understanding of the circumstances under which exclusion occurs. To this end, we explore household incomes in densification projects between 2012 and 2020 in the Province of Utrecht, the Netherlands, where urban development is traditionally strongly regulated through active land policy. At the same time, current shifts towards a more deregulated housing market make for an interesting case. Exceptional access to detailed cadastral and census data allows us to identify densification projects and assign them a median household income each. We investigate the influence of location and transformation process on household incomes through regression analysis and conduct qualitative case studies of projects whose median income was highly mispredicted by the regression model. This allows us to integrate non-quantified factors, such as land ownership and public policy interventions, in explaining such interesting cases. For the Province of Utrecht, our study

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confirms that while households in densification projects earn significantly more than their neighbours, the range of incomes in densification projects is large. Project characteristics such as centrality, neighbourhood status and transformation process explain only a small share of this variance. For cases where median incomes are much lower than predicted by the model, public land ownership, in combination with inclusionary zoning, is essential in ensuring housing affordability. Our approach highlights the necessity of supplementing densification policies with measures that secure affordable housing.

Keywords
housing affordability, land use policies, spatial justice, the Netherlands, urban densification

摘要
虽然世界各国政府都依靠紧凑的城市政策来减少城市增长带来的土地消耗，但最近的研究已经论证了致密化和住房负担能力之间存在潜在的需要权衡利弊的问题。有人担心，致密化会导致住房供应失衡，从结构上将低收入家庭排除在外。然而，很少有研究涉及致密化项目中家庭收入的变化，所以我们对排除发生的情况了解有限。为此，我们研究了荷兰乌得勒支省 2012 年至 2020 年致密化项目中的家庭收入，该省的城市发展历来通过积极的土地政策进行严格监管。与此同时，当前其房地产市场监管更加放松的转变也使得其成为一个有趣的案例。我们特别访问了详细的地籍数据和人口普查数据， 这使我们得以找到致密化项目，并为每个项目分配了一个家庭收入中位数。我们通过回归分析，研究地点和转变过程对家庭收入的影响，并针对一些中位收入被回归模型严重错误预测的项目，对其进行定性案例研究。这使我们能够整合非量化因素，例如土地所有权和公共政策干预，来解释这些有趣的案例。对于乌得勒支省，我们的研究证实，虽然致密化项目家庭的收入明显高于其邻居，但致密化项目家庭之间收入跨度很大。中心性、街区地位和转变过程等项目特征只能部分解释这种差异。对于收入中位数远低于模型预测结果的情况，公共土地所有权与包容性分区相结合，对于确保住房负担能力至关重要。我们的方法强调了有必要采取措施来保障经济适用的住房，以此来补充致密化政策。

关键词
住房负担能力、土地使用政策、空间正义、荷兰、城市致密化

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Introduction

Urban densification is considered key to combat land take and urban sprawl. Therefore, governments globally have imposed restrictions on land supply for construction, concentrating urban development within existing built-up areas. While densification, or infill development, is generally regarded as a viable approach to sustainable urban development, concerns centre around its connection to social sustainability, especially housing affordability (Teller, 2021). Although earlier studies acknowledge the potential benefits of densification, such as intensified interactions and improved access to public transport and job offers (Ahlfeldt and Pietrostefani, 2017; Burton, 2000), researchers stress the risk of densification creating a housing offer that deliberately excludes low-income households (Debrunner et al., 2020; Rérat et al., 2010).

The perceived risk of exclusion is strongly associated with gentrification as densification projects take place in existing neighbourhoods. Considerable studies have shown how former working-class neighbourhoods have been redeveloped into upscale areas, diminishing housing affordability in densifying

neighbourhoods (Cavicchia, 2021; Moos et al., 2018). Such exclusionary effects have been found regarding income, education level, migration background and age (Cavicchia and Cucca, 2020; Moos, 2016; Nachmany and Hananel, 2023). The fact that densification seemingly caters to young, highly educated and small households appears to do little to stop young families or older people from moving to or remaining in peripheral, low-density detached housing (Bromley et al., 2005; Moos, 2016; Steinacker, 2003).

While these insights have raised awareness of potentially negative social trade-offs of densification, the factors influencing the relationship between density and affordability appear largely unexplored. For instance, the age of the housing stock, city size and polycentricity can impact the effect of densification on income segregation (Garcia-López and Moreno-Monroy, 2018; Pendall and Carruthers, 2003). Effects differ between brownfield redevelopments, the direct replacement of social housing blocks and housing subdivisions (Bibby et al., 2021; Troy et al., 2017). Additionally, there are differences between local governments regarding the degree to which they combine densification with the goal of attracting higher-income households (Quastel et al., 2012). Such land policy factors should receive greater attention when considering the conditions for achieving urban densification while maintaining an inclusive housing supply (Cavicchia, 2021). Therefore, approaches that combine empirical insights on spatial processes and land policy interventions (Jehling and Hecht, 2022) are highly promising to describe and explain the social effects of densification.

Against this backdrop, this paper aims to develop and test a novel approach to explain the variation of household incomes across densification projects, asking: *What is the distribution of household incomes across densification projects at the city-region level? How do project location, transformation type, land ownership and planning interventions affect household incomes?* We argue that household income constitutes a promising indicator to operationalise our interest in studying the social effects of densification, as it allows us to analyse housing offers across renting and property markets and focuses on those who live there. Access to exceptionally detailed Dutch income, building and land use data allows us to assign median household incomes to densification projects between 2012 and 2020.

Following a neo-institutional approach, we understand densification outcomes as resulting from the interplay between institutions and actors' strategies. Therefore, we perform a regression analysis tracing the effect of transformation type and location on household income. Then, following a multi-method approach (Seawright, 2016), we qualitatively examine interesting cases – namely projects where the model vastly mispredicted household incomes. This allows us to include further causes, such as landowner strategies or public policy interventions (Jehling et al., 2020).

The spatial level of the province offers a city-regional perspective with sufficient projects for statistical analysis. It is simultaneously small enough for in-depth qualitative analysis in a comparatively homogeneous regional housing market. The Netherlands offers an interesting planning context to study densification. The efficient use of scarce land has been a central tenet of Dutch land use planning in various national spatial planning policy documents. It was further solidified with the introduction of the Ladder of Sustainable Urbanisation in 2012, prioritising developments within existing urban areas. In addition, against a backdrop of housing market deregulation and the shift away from municipal land ownership, the Netherlands provides an interesting case for international observers, particularly

in exploring the relationship between public land ownership and housing affordability in densification projects (Claassens et al., 2020; Musterd and Ostendorf, 2021). The remaining sections of the manuscript encompass the theoretical framework, methodology, results, a discussion of the findings and concluding remarks.

Explaining housing offers through property rights and public policies

Following a neo-institutionalist approach, we understand housing offers through densification as an outcome regulated by property rights and public policies. These two sets of rules determine how actors can gain access, use or exploit resources such as land or housing. They, therefore, enhance or restrict actors' use interests (Gerber et al., 2018). Public policies, in particular planning and housing policies in the context of this research, aim to regulate the behaviour of landowners to solve issues in the distribution of housing (Knoepfel et al., 2007). On the other hand, property rights aim to protect the individuals' interests from interference from the state. The two sources of formal rules and the appropriation strategies of actors thus shape the housing outcomes in densification projects.

Property rights: Market forces influencing housing offers

Property rights enable actors to follow a market logic. Independently from public policy intervention, we expect the housing offer to reflect factors such as location, construction costs and developer strategies. Locational factors of a densification project encompass neighbourhood status, centrality and property prices. Since densification has been observed to occur predominantly in areas of high demand (where financial viability is given), it is also considered less affordable than other housing (Steinacker, 2003). As a form of risk management, developers mostly build similar to what already exists in the neighbourhood – except for gentrifying neighbourhoods where a large rent gap opens up possibilities to attract higher socio-economic groups (Kim, 2016).

In addition, construction costs vary between different kinds of densification projects. As an extreme example, subdividing a house into two or more apartments is less costly than transforming a brownfield. As brownfields may be contaminated, redevelopment can be expensive, time-consuming and risky (De Sousa, 2000). Thus, more low-income residents are expected to live in subdivisions than in brownfield redevelopments. Different groups of developers with different business strategies perform different kinds of densification projects. Some developers, expecting a direct return, build owner-occupied units that they can sell immediately (Rérat et al., 2010). As households in owner-occupied units are generally much wealthier than renters, this might lead to a higher average income in such projects (Arundel and Hochstenbach, 2020). Other investors, such as pension funds, are interested in long-term returns and incentivise the development of rental housing, also for the upper to middle class (Rérat et al., 2010), while individual, private landowners concentrate on subdividing and renting out smaller apartments (Bouwmeester et al., 2023). Thus, different types of developers may make the provision of certain housing offers more likely than others.

Public policies: The impact of planning interventions

Public authorities can intervene in private developers' property rights through public policies. Public policies can be defined as decisions by public authorities to resolve a politically defined collective problem (Knoepfel

et al., 2007). Thus, policy objectives constantly change as the understanding of collective problems evolves and political majorities shift. For example, through affordable housing policies, public authorities can try to steer developers to provide housing for low-income residents through the municipal building code, the provision of subsidies or negotiated land use plans (Debrunner and Hartmann, 2020). Contrarily, city authorities can implement policies to attract wealthier residents and increase social mixing (usually at the cost of lower-income households) (Lees, 2008; Uitermark et al., 2007). However, the effectiveness of public policies can be questioned. Debrunner and Hartmann (2020) find that even though planning instruments exist that could force investors to provide affordable housing, municipalities often do not apply these instruments. One major obstacle is that many planning instruments are relatively weak in front of well-protected property rights. Landowners are especially powerful in the context of densification projects. As land is scarce, public authorities depend on landowners to implement policies.

This section discussed variables that can explain differences in housing offers (and ultimately resident structures) between densification projects. In the following section, after explaining how we detect densification projects, we will present how the variables discussed above will be used in the further analysis of income variation across densification projects.

Methods and data

Identifying and describing densification projects at the province and neighbourhood level

We use information on former land use (t_0) and construction year to select housing units, that is, apartments in multi-family housing and buildings in the case of single-family housing, that were newly created at t_1. If the former land use was urban (i.e. no natural or agricultural land use), the housing unit is labelled as densification, otherwise as expansion. We further distinguish between the transformation processes 'urban green space transformation', 'brown- and greyfield transformation', 'densification in residential areas', 'densification that included the demolishing of existing housing units (redevelopment)' and 'soft densification' (Götze and Jehling, 2023). Contrary to 'densification in residential areas', which requires the construction of a new building, 'soft densification' only covers housing units that were created within existing buildings, for example, through the subdivision of apartments or transformation of offices, shops or attics (Bibby et al., 2020).

Each housing unit is assigned information on its residents, including age, household size, personal living space, education and household income. New housing units in spatial proximity are grouped into densification projects. We then analyse the distribution of socio-economic groups of the project compared to (1) all existing residents in the province and (2) existing residents in the respective neighbourhood.

Explaining the distribution of household incomes in densification

We employ multiple regression analysis to measure the effects of location and transformation type on household income distribution. This analysis is supplemented with qualitative case studies to examine the influence of land ownership and municipal intervention on median income.

Choice of the dependent variable and aggregation to projects. We take the median standardised household income in densification projects as an indicator for the dependent variable. Such standardised household incomes correspond to disposable incomes adjusted for differences in household size and composition (Statistics

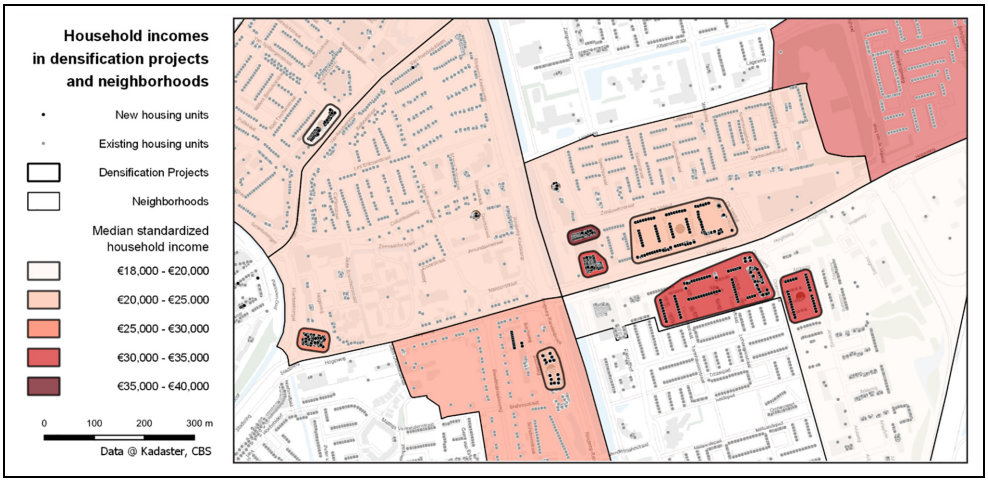


Figure 1. Median standardised household incomes in densification projects and neighbourhoods, 2019.

Netherlands, 2018). Compared to housing prices, incomes represent directly who lives in a housing unit and cover both tenants and owners. This approach also considers that households in central locations may have the capacity to allocate more funds towards rent due to reduced reliance on car ownership for commuting (Aurand, 2010; Xiao et al., 2016). Since it is our aim to cover all socio-economic groups living in densification projects, we also keep students and retirees in the dataset. This allows us to find potential student dorms or retirement homes that have been constructed. Robustness checks indicate that students and retirees negligibly affect the significance and coefficients of the regression model (Online Supplemental Figures S01 and S02). While the approach is well-suited for the aim of this article, it must be stressed that household income does not directly reflect affordability, as it ignores the share of income required for housing.

Densification projects are formed by aggregating ten or more households. This has several advantages. First, income variance within projects is often high, and reducing the information to a single median value per project reduces this noise. Second, we aggregate into projects to reduce spatial autocorrelation because the similarity of incomes among households in the same building can violate the assumption of independence in regression analysis. This can potentially distort the relationships measured in the model. A disadvantage of this decision is that developments with less than ten households (often soft densification) fall out of the regression analysis. To cover their importance in densification (Bibby et al., 2020), they are still considered when measuring the distribution of standardised household incomes across development types.

To group housing units into densification projects, we use a density-based clustering algorithm (DBSCAN). This algorithm clusters data points based on a maximum point-to-point distance (Eps) and a minimum number of points that can form a cluster (MinPts) (Ester et al., 1996). We use a maximum point-to-point distance of 35 m with a minimum number of ten units per cluster (Figure 1).

Multiple linear regression analysis based on actors' interests and policies. Multiple linear regression analysis estimates the effect of demand

and construction costs on standardised household incomes. We use the following predictors: transformation process, centrality in 2011, neighbourhood income 2011 and neighbourhood income change 2011–2019. The centrality is measured as address density within a circle of 1 km^2 around each address in a neighbourhood (Van Leeuwen and Venema, 2023). Neighbourhoods are defined following the delineation of Statistics Netherlands. Neighbourhood income 2011 and income change 2011–2019 represent their status and dynamics, indicating attractiveness for developers. The indicators are based on the median standardised household income per neighbourhood in 2011. Neighbourhood income changes, then, depict the difference between a neighbourhood's median standardised income in 2019 and 2011, corrected for inflation. Residents in newly constructed addresses are filtered out of the calculation to avoid simultaneity bias (i.e. newcomers lifting average neighbourhood income). For the same reason, the variables 'centrality' and 'neighbourhood income' reflect measurements from 2011, before densification happened.

Qualitative case study analysis. We select projects where predicted income differs most from real income, that is, residuals exceeding ±€10,000 (following Garcia-Lamarca et al., 2021). Analysing such deviant cases is valuable for hypothesis building since it allows for identifying further causal relations that explain densification outcomes (Lieberman, 2005; Seawright, 2016). To analyse these cases, we collected and analysed legally binding documents, such as land use plans, visions and official municipal decisions, as well as non-binding documents, such as meeting minutes of municipal councils, newspaper articles and strategic documents.

Data sources and data access

Housing units with construction year and surface area are retrieved as point data from the Dutch cadastre. Statistics Netherlands provides publicly accessible vector data on land use and neighbourhood aggregated data on address density (i.e. centrality). Access to non-public household-level microdata on income, age, household size and education was granted by Statistics Netherlands. To calculate neighbourhood income, we aggregate income data to pre-defined neighbourhoods.

Out of 57,633 housing units that were newly registered in the cadastre between 2012 and 2020, 38,376 are identified as densification (the remaining units as expansion). We aggregated these 38,376 housing units into 436 densification projects that were then used in the regression analysis. Of the 38,376 housing units, 5,437 are not part of densification projects and were thus excluded from the regression. In the Online Supplemental Materials, you can find summary statistics (Supplemental Table S01) and a correlation matrix (Supplemental Figure S03) for the variables that enter regression analysis.

Results

Distribution of standardised household incomes

With €30,800, the median standardised household income in densification projects (excluding soft densification) is slightly higher than the Province median of €30,700 and considerably lower than in expansion areas (€35,700) (Figure 2). Compared to existing households in the same neighbourhood, the newcomers' incomes lie on average €3,700 above the neighbourhood median.

The transformation types of soft densification and redevelopments in residential areas show the lowest incomes. In contrast,

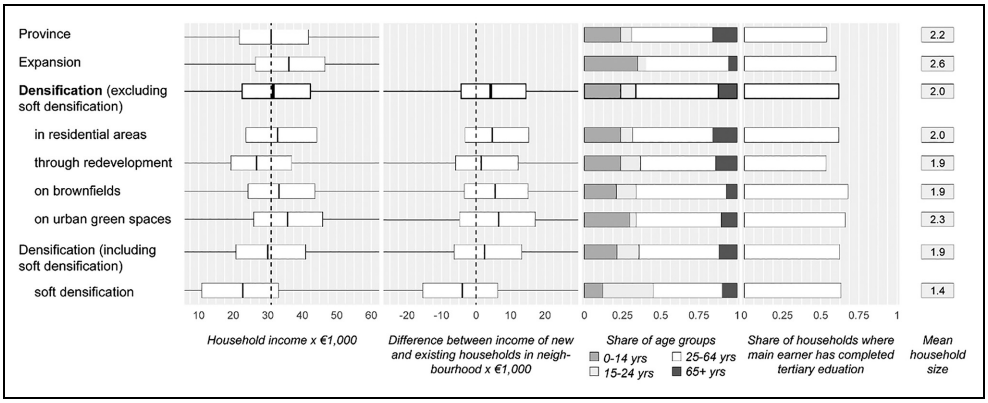


Figure 2. Socio-economic characteristics of residents at t_1 grouped by transformation process. See for classification of education levels: https://www.cbs.nl/en-gb/our-services/urban-data-centres/labour-and-income/education-level#:~:text=Education%20levels%20can%20be%20broken,vocational%20education%20(MBO)%2C%20higher.

the transformation of urban green shows the highest incomes, comparable to those observed in expansion areas. Consequently, if we include soft densification, incomes in densification projects move below the province median but are still, on average, €2000 higher than the neighbourhood median.

Only in the case of soft densification projects and residential redevelopments do newcomers earn less or almost the same as the existing residents in the neighbourhood. At the same time, projects in these categories that together make up 40% of all densification projects in the analysis occur on average in neighbourhoods with low median incomes of respectively €22,400 (redevelopment) and €21,800 (soft densification) (Online Supplemental Table S02).

Households in green space transformations resemble those in expansion areas regarding household size and share of children. In contrast, households in other forms of densification projects are comparatively smaller than the province's mean. Households in green space transformations even enjoy, on average, 5 m^2 more living space than those in expansion areas and 8 m^2 more than households in brownfield transformations (Online Supplemental Table S03).

The highest share of main earners with tertiary education is reached in brownfield redevelopments. Also, residents in soft densification projects have, to a large degree, a completed tertiary education, distinguishing them from residents in residential redevelopment projects (i.e. demolish-rebuild) with whom they share low-income levels. In addition, soft densification projects show a remarkably large share of residents between 15 and 24 years of age, approximately five times higher than the provincial average.

Median standardised household income in densification projects of ten or more households – explained by regression analysis

The median income in densification projects is significantly and positively related to centrality (i.e. address density) and neighbourhood incomes (Table 1). Of the densification processes, residential redevelopments and

Table 1. Regression coefficients, standardised household income in infill projects.

Dependent variable	Median standardised household income 2019
Transformation process variables	
Redevelopment in residential areas	−2544***
Soft densification	−6523***
Brownfield transformation	2444**
Urban green space transformation	2716*
Neighbourhood variables	
Centrality 2011	1.04***
Median Neighbourhood Income 2011	0.68***
Change in median neighbourhood income 2011-2019	0.39**
Intercept	13,160***
R^2	0.22
Adjusted R^2	0.21

***Significant at 1%. **Significant at 5%. *Significant at 10%.

soft densification show a significant negative, and transformations on urban green spaces and brownfields show a significant positive difference to the null hypothesis of densification in residential areas. The adjusted Pearson correlation of the model is low at $R^2 = 0.21$. However, the residuals are normally distributed and show a low Moran's I value for spatial correlation (Online Supplemental Figures S04 and S05).

Concentrating on the residuals, we further examine the relationship between densification projects' income and neighbourhood attributes. We focus first on projects with household incomes that we consider rightly predicted by the model (residuals of ±€5000) and later explain projects where incomes have been greatly mispredicted by the model (residuals of ±€10,000). For a fifth of all projects, the regression model over- or underestimated median household incomes by over €10,000 (Figure 3).

Projects with correctly predicted household income

We start by investigating what characterises projects with a rightly predicted median standardised household income in the lowest quartile (<€23,000). All of the 14 projects were instances of 'densification that included the demolishing of existing housing units (redevelopment)' or 'soft densification'. Five are located in most central areas (top quartile), but none are in the highest income neighbourhoods (top quartile). Still, in nine projects, newcomers earn less than their neighbours. There are also examples of low-income households moving to strongly gentrifying neighbourhoods, but only through soft densification. The group of rightly predicted projects with the highest median incomes (top quartile, >€36,000) is made up almost entirely of brownfield and urban green space transformations and infill in residential areas. Only one project, situated in a top-income-quartile neighbourhood, was created through redevelopment. Many high-income projects are in the most peripheral regions (lowest quartile). Only one project was constructed in a bottom-income-quartile neighbourhood, and four were built in neighbourhoods with a below-median income in 2011 (<€23,000). One of them, a sports field transformation in the city of Utrecht, produced rowhouses with a median income of €37,000.

Projects with a mispredicted household income – case studies

To understand why the median incomes of certain projects have been mispredicted, we need to understand the policy context in which densification occurs in the Netherlands. Dutch planning authorities have traditionally had a strong influence on spatial developments and the housing market. Land uses have been tightly coordinated through the

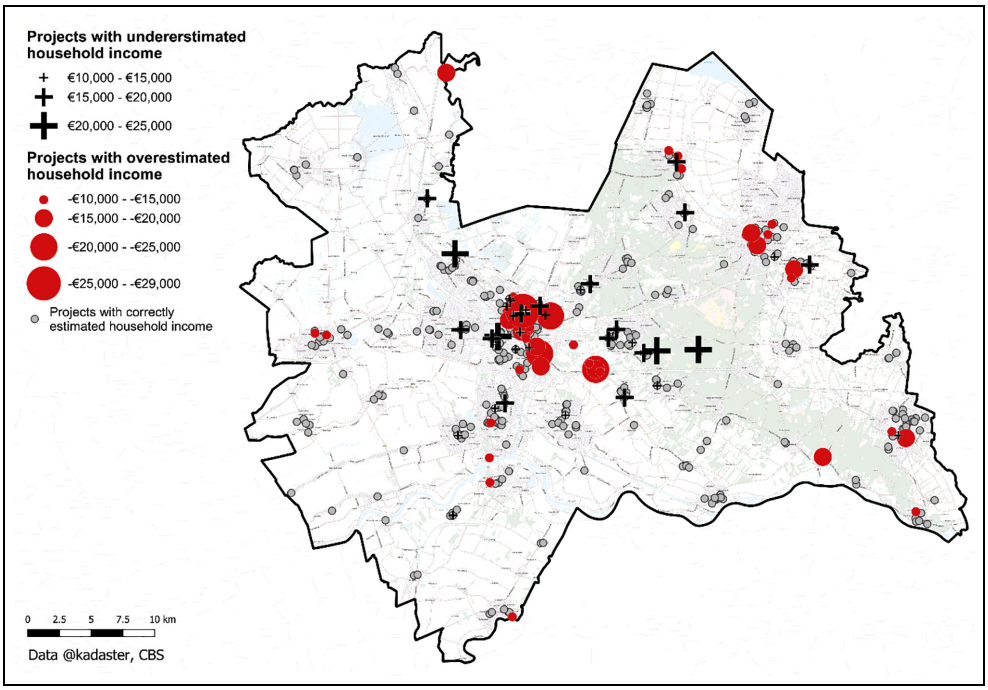


Figure 3. Under- and overestimated densification projects and rightly predicted projects.

national government and the use of active land policy. After WWII, housing associations (not-for-profit actors) played an important role in rebuilding efforts. As a result, social housing was widely available for people of every socio-economic status (Buitelaar, 2010). Housing associations still hold a sizeable percentage of ownership in early post-war neighbourhoods (Priemus, 2006). However, new housing policies implemented after the crisis have led to a declining share of stock from 40% in 1990 to about 29% in 2022 (CBS, 2023). In addition, regulatory changes have limited housing associations' ability to acquire land as they can only hold it for five years, and extra taxes on social rent income have created financial pressure (van Gent and Hochstenbach, 2020).

These changing policies are part of a general shift in ideas about the state's role in urban development and housing construction. On a municipal level, this is most obviously characterised by the shift away from active land policy after the global financial crisis when municipalities suffered big losses on land development. Instead, local planning authorities take a more facilitating role and are expected to provide room for initiatives from the private sector (van Oosten et al., 2018). For most redevelopment projects, local planning authorities now renegotiate part of the relevant land use plan with the developer, making it more challenging to enforce inclusionary zoning. Still, municipalities have some instruments available to steer housing construction. In the region of Utrecht, some municipalities have included a rule in the land use plan that stipulates that a certain percentage (often 30%) of new construction needs to be social housing.

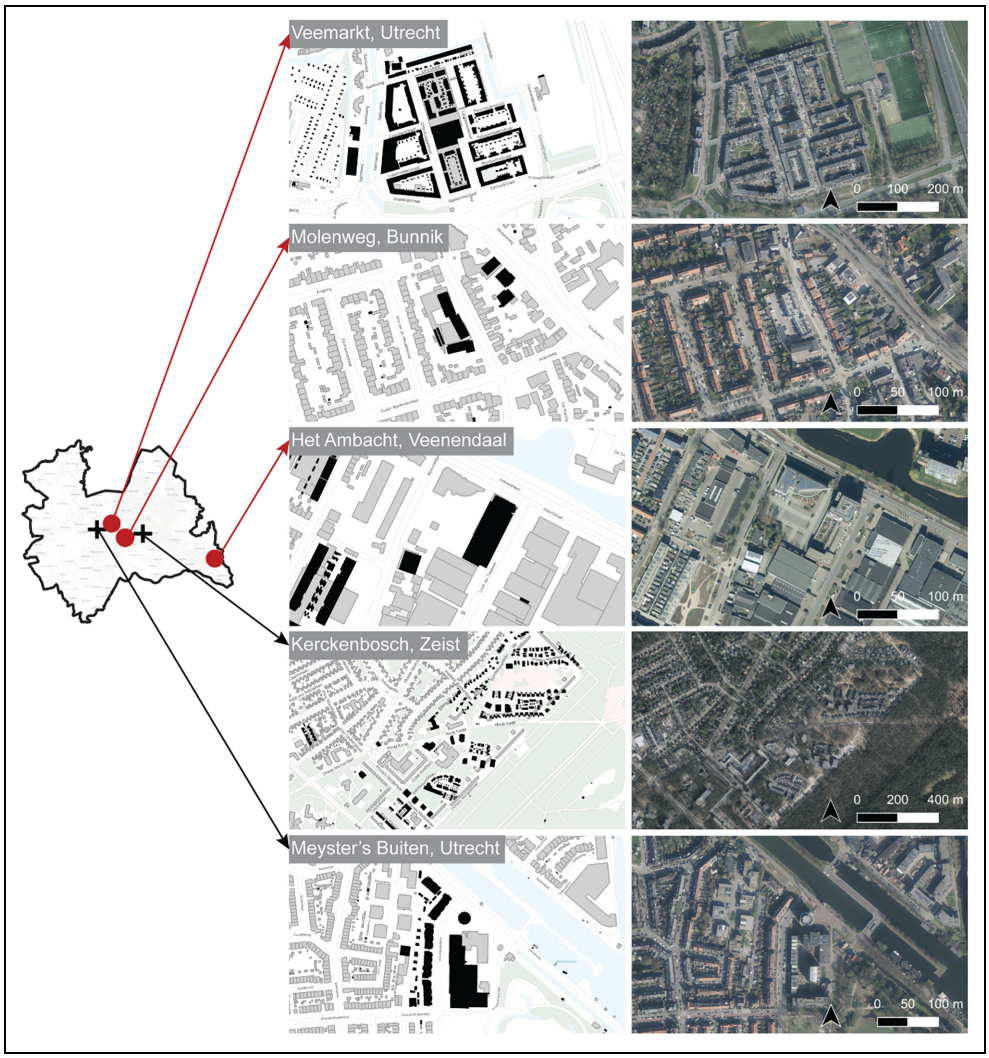


Figure 4. Case study. Data @Kadaster.

Projects with overestimated household income. Many of the projects with overestimated incomes are characterised by the fact that they were realised on (once) publicly owned land. A good example is a large redevelopment project in the east of Utrecht city called Veemarkt (Figure 4). Through public tenders, the municipality could implement objectives and ambitions such as sustainability. Another objective was to provide 40% social rent or affordable owner-occupied housing (Municipality of Utrecht, 2013). Since the municipality of Utrecht had made agreements on fixed land prices for plots on which social rent would be developed, these plots did not have to be given out through a tender but were negotiated among different housing associations in Utrecht (Municipality of Utrecht, 2011a).

Another project with a high overestimation – assisted living apartments for people with a

disability – is located in a smaller town called Veenendaal and concerns the redevelopment of a plot in the industrial park Het Ambacht. The industrial park is one of the municipality's main redevelopment areas. In this project, land ownership was in the hands of a private developer and a housing association, who purchased the land because of the planned redevelopment of the industrial park. The two parties worked together to realise a residential care complex (Patrimonium Woonservice, 2016). In response to the initiative of the two parties, the municipality implemented a new land use plan in 2013, allowing for a change in function (Municipality of Veenendaal, 2013). In this case, incomes in the project are lower than expected because of the land ownership by a non-profit housing association.

Other projects with overestimated incomes cannot be explained through public land ownership, ownership by a non-profit private actor nor through qualitative targets in public policies. An example is the Molenweg project in the small town of Bunnik. This neighbourhood is dominated by owner-occupied housing, but a former industrial site was transformed into rental apartments. As stipulated by the housing vision, the municipality has a housing shortage in the higher-intermediate segment (€1,000– €1,200/month) for the elderly who want to move to more age-appropriate housing (Municipality of Bunnik, 2018). While initially, the project developer did research the possibility of realising single-family housing in this location, the project developer and the local planning authority agreed that 24 rental apartments would be constructed in the higher-intermediate segment in 2019. This option was 'more attractive because of the public housing task and market demand' (Van Wanrooij Projectontwikkeling, 2019: 1). The case shows that municipalities can sometimes negotiate the construction of comparatively affordable apartments with the developer.

Projects with underestimated household income. The project with the greatest underestimation of income achieved a median standardised household income of €69,000 (€24,000 above the modelled value). Residents thus belong to the 3% with the highest incomes in the Netherlands. The spacious single-family units mimic the style of the popular surrounding 1930s neighbourhood *Oog in Al*, 1.5 km from Utrecht central station. Marketed as 'royal mansions' (Blauwhoed, 2011), the concerned row of houses is part of the larger transformation project *Meyster's Buiten*. In our analysis, however, the project was cut in two, separating the mansions from the denser, publicly owned, mixed-use development to the east (median income: €45,000). Meyster's Buiten is a collaboration between the municipality of Utrecht and two private developers. In 2011, after a fire destroyed most of the factory buildings and during the onset of the financial crisis, the three parties had to adapt the original development plan to increase profitability (Wong, 2015). In this context, planned apartment buildings were replaced by single-family units (Municipality of Utrecht, 2011b).

Another project with a highly underestimated median income concerns the redevelopment of social housing blocks from the 1960s at the forest edge in the neighbourhood of Kerckebosch, east of Utrecht. Here, on land formerly owned by the Municipality of Zeist and a social housing association, approximately 700 social housing units were replaced by 1000 new units, of which 55% are social housing (Bosoni, 2020). Green wedges intersect the new building groups that are again registered as individual projects rather than a contiguous one. Correspondingly, while many building groups show the expected low median household incomes, one was underestimated by €23,000. This can be explained by the financing scheme of the redevelopment

project. In this scheme, the construction of social housing during later construction phases is financed through the sale of condominiums in earlier phases. In this project, it was argued that changed circumstances after the financial crisis made it necessary to replace planned apartment buildings with more profitable single-family units (Municipality of Zeist, 2014).

Discussion

For the Province of Utrecht, our findings show that, while households in densification projects, on average, earn more than their neighbours, household incomes vary significantly across projects. Project characteristics, such as location and transformation process, only explain household incomes to a small degree. In many projects where the newcomers' income deviates a lot from expectations, municipalities were able to steer project outcomes through active land policy.

While supporting earlier studies showing that households in densification projects earn more than average (Cavicchia, 2021; Rérat et al., 2010), our study additionally explores what factors explain differences in household income between densification projects. Not surprisingly, projects in more central locations and higher-income neighbourhoods also show higher median household incomes. However, even centrally located projects in moderately wealthy neighbourhoods can show below-average income levels, given they are soft densification or redevelopment projects.

In the case of soft densification, the resulting apartments (or rooms) are significantly smaller than those in their surroundings (Götze and Jehling, 2023). It is an inexpensive strategy of individual property owners in response to the high demand for housing in city centres. In the case of Utrecht, this practice is sometimes mentioned in the context of student rentals (Bouwmeester et al., 2023). This is supported by the high shares of young adults in such projects, reflecting the rising popularity of high-density living among this age group (Moos, 2016; Rérat, 2019). At the same time, soft densification projects show comparatively high shares of residents with completed tertiary education. Both findings point to the need to include measures of age and education next to income in future studies of residential segregation (Boterman et al., 2021).

For redevelopments (i.e. demolition–rebuild projects), low median incomes are likely explained by the fact that this transformation type is performed chiefly on rental housing blocks, of which, in the Netherlands, 70% are owned by non-profit housing associations. In this case, however, the redevelopment happens at the cost of existing affordable housing units and is often accompanied by the eviction of previous residents (Musterd and Ostendorf, 2021), additionally supporting concerns about gentrification effects. Further studies should, therefore, also employ socio-economic data of those who are displaced through densification. In general, densification predominantly occurs in less affluent but well-located areas, where large rent gaps make it profitable (Kim, 2016), while more affluent communities successfully prevent densification through their property rights (Charmes and Keil, 2015; Touati-Morel, 2015). This location bias and intervention in vulnerable neighbourhoods sets densification apart from greenfield development, which, while also targeting higher-income households, takes place on former uninhabited land.

Still, the location and transformation process explain only a small share of the variance in median household incomes, as reflected in the relatively low fit of the regression model comparable to earlier studies (Garcia-Lamarca et al., 2021; Steinacker, 2003). Acknowledging that planning and housing policy in the Netherlands intervenes

in housing markets quite significantly, this was to be expected since essential factors, such as public land ownership and planning interventions, were not covered by the model. Consequently, we added a qualitative case-based explanation for interesting cases where the model strongly mis-predicts median household incomes.

The case studies of projects where median household incomes were strongly overestimated reaffirmed the important role of municipal land ownership in providing affordable housing. Non-profit housing associations rely on land transfers from municipalities because they cannot usually compete with market players. This has to do with continually tightening regulations that make it increasingly difficult for housing associations to acquire land. The Housing Act of 2015 introduced stricter regulation concerning the involvement of housing associations in the non-social rent sector and their ability to speculate on future land developments. Simultaneously, it has become possible for private actors to supply social housing. With housing associations thus being limited in their ability to acquire new land, they have become more dependent on private developers to sell them newly constructed buildings. Alternatively, they can increase their housing stock through the densification of their existing plots (demolition–rebuilt). Still, as the case in Veenendaal shows, housing associations can sometimes secure land ownership in redevelopment cases without any public land ownership. In these cases, they have to act according to a financialised logic, using their equity or selling older housing stock to compete with commercial actors (Aalbers et al., 2017; Buitelaar, 2010). Our case study of underestimated projects has shown how both municipalities and housing associations have financed the construction of affordable housing by selling expensive condominiums within the same project. This was partly revealed through the applied approach to aggregate densification projects, which splits larger projects with cross-financialisation into separate projects.

In addition, examples among projects with both over- and underestimated incomes showed that the financial crisis of 2007–2008 and the following drop in construction until 2014 made it difficult for municipalities to implement social housing quotas. Only recently, in the wake of an overheated housing market and, subsequently, rising house prices, did municipalities in the province start applying quotas to new construction projects. These quotas are likely to impact household incomes in densification projects but are not reflected yet in the data used in this paper.

The presented approach showed great potential for exploring the factors that influence household income in densification projects. Crucially, highly detailed income and building data allowed for a precise distinction of densification projects and their residents from their surroundings, covering a complete city region (Götze and Jehling, 2023; Jehling et al., 2020). In addition, combining regression analysis and qualitative case studies proved helpful in highlighting interesting cases (Seawright, 2016). While using the indicator ‘household income’ had the advantage of covering both tenants and owners, it must be stressed again that it is not a direct representation of housing affordability.

Conclusion

Against the backdrop of concerns regarding the potential exclusion of low-income households due to urban densification, this study set out to explore factors accounting for differences in median household incomes across densification projects. Access to microdata allowed us to distinguish newcomers from existing residents, making it possible to calculate median household incomes for

individual densification projects. In addition, by combining multiple regression analysis with case studies of mispredicted cases in a multi-method approach, we can consider both quantitative factors (location and transformation type) and qualitative factors (land ownership and public policy interventions) in explaining income across densification projects.

While our findings for the Province of Utrecht have confirmed that households in densification projects earn more than their direct neighbours, we have also observed considerable differences between projects. Factors such as centrality, neighbourhood status and transformation type explain household incomes only to a small degree, leaving 80% of the variance unexplained. Public land ownership has proven powerful in providing housing for lower-income households in the projects that we examined qualitatively. However, such case studies have also shown the vulnerability of financing schemes, even on publicly owned land, where the provision of affordable housing depends on the profitable sale of owner-occupied housing within the same project. Potential for further research lies in including measures of age and education, as well as displacement connected to various forms of densification. Our contribution shows that the relationship between density and housing affordability is inherently political, shaped by decisions about who should have access to land and housing.

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Supplemental material

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